D. DHARSHANA
192524243
CSA051

CO1 - AT2 - CONCEPT MAPPING

SECURITY ATTACKS

Attempts to access, alter steal or disrupt information of system resources.
Ex: phishing, Data tampering, DDoS

creates

may lead to

THREATS

Potential dangers to systems, data or users.
Ex: password theft, unauthorized modification, service disruption

can lead to

RISKS

Possibility of loss or damage due to threats.
Eg: financial loss, incorrect records, website downtime

SECURITY GOALS

Protect information through the CIA Triad.

CONFIDENTIALITY

Prevents unauthorized disclosure of information
Ex: Encryption protects bank details

INTERGRITY

Prevents or detets unauthorized changes to data.
Ex: Hash verification detects altered files.

AVAILABILITY

Ensures authorized users can acess services & data.
Ex: DDOS protection keeps website online.

SUBSTITUTION TECHNIQUES

Replace each letter/symbol with another based on a key or rule

CAESAR CIPHER

- Key usage: Single integer key (shift value 0-25)
- Complexity: low
- Security strength: Weak

Example: Shift 3
Plaintext: HELLO
Ciphertext: KHOOR

MONOALPHABETIC CIPHER

- key usage: 1-to-1 Substitution (26 letter mapping
- Complexity: Medium
- Security strength: Moderate

Example: Mapping
A → X, B → Q, C → G...
Plaintext: ABC
Ciphertext: XQG

VIGNERE CIPHER

- Key usage: keyword (repeated to match plaintext length
- Complexity: High
- Security strength: stronger than caesar + monoalphabetic

Example: key: KEY
Plaintext: ATTACK
Cipher text: KFFBOP

HILL CIPHER

- Key usage: Matrix key (eg: 2x2, 3x3)
- Complexity: High
- Security strength: Strong (against simple attacks

Example: 2x2 matrix key

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \pmod{26}$$

MOST SECURE TECHNIQUE

Hill cipher is the most secure among the four because it uses a matrix key, which creates complex transformation of the plaintext, making it harder to break with simple frequency analysis or substitution attacks

CLASSICAL ENCRYTION SCHEMES

SUBSTITUTION TECHNIQUES

Replaces each letter or symbol with another based on a key or rule

CAESAR CIPHER

- Data Structure: Same length, letters replaced (A → D, etc).
- Security level: Low
- Example: HELLO → KHOOR

MONOALPHABETIC CIPHER

- Data structure: same length, letters replaced using a fixed key (mapping).
- Security level: Medium
- Ex: HELLO → XEBKQ

SIMILARITIES

- Both are classical methods.
- Both use key (or rule).
- Both aim to make data unreadable without the key.

TRANSPOSITION TECHNIQUES

Rearranges the positions of letters / symbols without changing them

RAIL FENCE CIPHER

- Data structure: same length letters rearranged in a zig-zag pattern.
- Security level: Medium
- Ex: HELLO → HLOEL (2 rails)

COLUMNAR TRANSPOSITION

- Data structure: same length rearranged by columns.
- Security level: medium to high.
- Ex: HELLO → EOLHL (key: 2 4 3 1 5)

Which is more vulnerable?

- Caesar cipher - most vulnerable (very simple).
- monoalphabetic cipher: Vulnerable - vulnerable to frequency analysis

MOST SECURE TECHNIQUE

Columnar Transposition

It is more secure than the others because it rearranges the data in columns using a key, making frequency analysis much harder

Why?

- use a keyword / column order
- Breaks simple patterns in the text.
- Requires more effort for an attacker to find the orf arrangement

STEGNOGRAPHY

Data Hiding

- Hides secret data within a cover file (image/audio/video/text)
- Main goal : keep the existence of the message secret.
- usually invisible to the human eye.

Watermarking

- Embeds an identifier mark (logo text, image) into the media.
- Main goal : prove ownership. authenticity or integrity.
- can be visible or invisible (but usually detectable).

Survivability

- The ability of the hidden data / watermark to remain intact even after modifications (noise, cropping, etc.).
- Ensures the data can still be recovered or verified.

How they are related.

Relationship

- Data hiding uses stegnography to conceal the message
- Watermarking is a special type of data hiding with specific purpose
- Survivability ensures the hidden data / watermark stays intact + it recoverable

Contribution to secure communication

- Data hiding → keeps the message secret from attackers
- Watermarking → proves the data is genuine (integrity + authenticity)
- Survivability → ensures the message or watermark is not lost or easily altered (reliability)

Real - World Applications

- Data hidden text in an image (eg.. secret message in an photo)
- Watermarking → copyright mark in digital images / videos
- Survivability → fragile watermark in medical images